

1. Kinematics of a particle moving in a straight line or plane

What students need to learn:

Motion in a vertical plane with constant acceleration, e.g. under gravity.

Simple cases of motion of a projectile.

Velocity and acceleration when the displacement is a function of time.

The setting up and solution of equations of the form $\frac{dx}{dt} = f(t)$ or $\frac{dv}{dt} = g(t)$ will be consistent with the level of calculus in C12 and C34.

Differentiation and integration of a vector with respect to time.

For example, given that $\mathbf{r} = t^2\mathbf{i} + t^{3/2}\mathbf{j}$, find $\dot{\mathbf{r}}$ and $\ddot{\mathbf{r}}$ at a given time.
